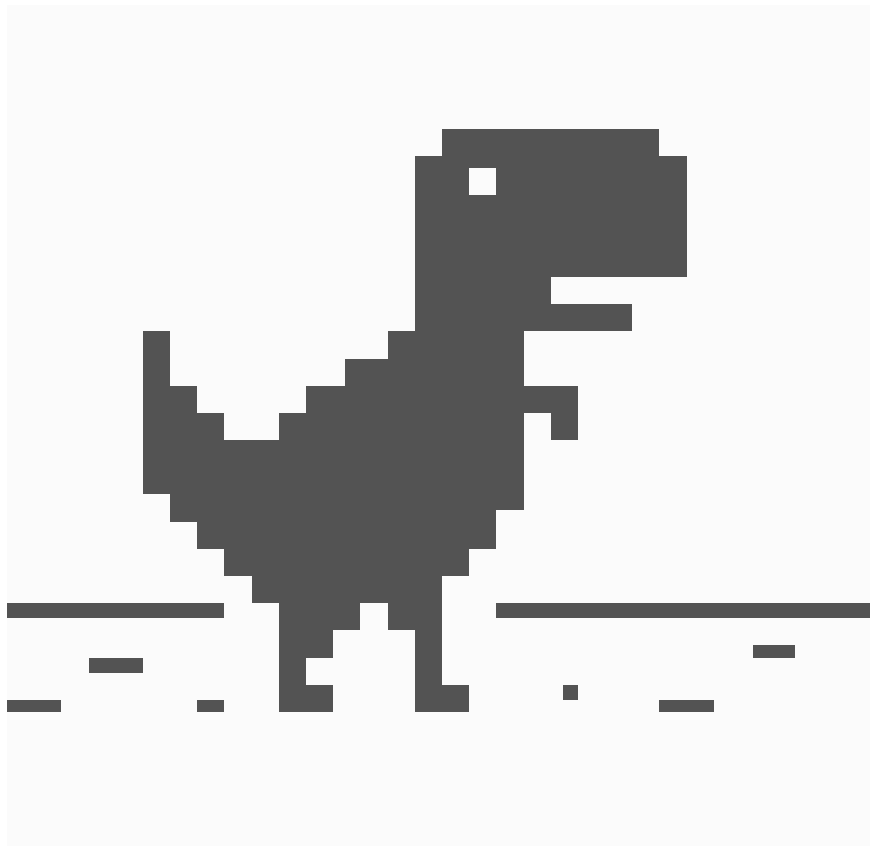


Lab Manual



[ROARS](#)

George Mason University

This manual serves as a guide for prospective and current members of [Roars Lab](#). It outlines the lab's policies, procedures, and expectations to ensure a productive and collaborative research environment.

Updating the manual: This document is a living document in Roars Lab's Github repo at

<https://code.roars.dev/phd-cs-us/lab-manual.typ>

Lab members, who all have write access, are encouraged to provide feedback and suggest changes to improve its clarity and usefulness.

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1. Publication

2. Writing Styles and File Organization

Everyone has their own way of organizing files and writing papers. Below is my approach that has worked well for me over the years. If you're my student and write papers with me, I expect you follow this approach so that we can be on the same page and work efficiently together.

Note that if we collaborate with others, we can adapt how we write and organize files as needed. Typically, the main group lead will decide how to organize things and the rest of us will follow their lead. For example, if we lead a paper, then we will follow my approach below. But if we collaborate on a paper led by others, then we will follow their approach.

2.1. File Organization

1. Use *Git control* (e.g., *Github*) for version control. This is essential for managing changes to the paper, especially when collaborating with others.
 - Even if we use something like *Overleaf*, link it to a Github repo.
2. *Keep it simple*. I prefer a very simple file organization structure for writing papers. As few files and directories as possible.
 1. `paper.tex`: a *single* TeX file for the entire paper.
 - Some might prefer using multiple files / directories (e.g., `intro.tex`, `eval.tex`, `sections/related.tex`). But I find it easier to just use a single file because I can easily search and replace for things and I don't have to switch between files, which interrupts my flow.
 - Even when sharing or collaborating with others, where conflict edits can arise, a single file still works well because Git is pretty good at resolving conflict issues.
 - I use this approach for all projects, including proposal (`proposal.tex`) and presentation (`slides.tex`) files.
 - Sometimes I append things to the end of the file to indicate the venue (e.g., `paper-cacm26.tex` for a paper submitted to CACM). But the main idea still remains: an entire paper in a single file.
 2. `paper.bib`: A single `bib` file for bibliography references.
 - Using a *single* `bib` file makes it easier to manage and search for references. Others might prefer multiple `bib` files (e.g., `refs1.bib`, `refs2.bib` for different sections).
 - My collaborators sometimes put in their own `bib` files.
 - Do not symbolically link to a shared `bib` file (e.g., `~/myrefs.bib`) because it can cause issues when sharing the paper with others (e.g., for review or collaboration), and also if you update the shared `bib` file, it can cause issues for other work that also uses that `bib` file. Instead, just copy the `bib` file to the paper directory and use it there and make things self-contained.
 - `bibentry` conventions: use `authorYearFirstWordOfTitle` as the `bib` entry key, e.g., `duong2020verifying` for a paper by Duong in 2020 with the title "Verifying ...". This makes it easier to remember and use the `bib` entries in the paper. Note that Google Scholar also uses this convention when you export `bib` files, so it is easy to copy and paste from Google Scholar to your `bib` file.
 - If there are multiple papers by the same author in the same year, add a number to the end of the `bib` entry key, e.g., `duong2026verifying`, `duong2026verifying2`, etc.
 3. `figures/`: A directory (folder) containing all figures for the paper.

- If there are just a few figures, I might just put them in the same directory as the paper file. But if there are many figures, I prefer to keep them in a separate directory.
- Same as the bib file, do not symbolically link to a shared figures directory (e.g., `~/myfigs/`) because it can cause issues when sharing the paper with others.

2.2. Writing Order for Research Papers

I often write papers in the following order. For a concrete example, see the [Dynaplex paper](#) published at OOPSLA'21.

1. **Overview** section (Section 2).

- Start with this section, which should contain an **illustrative or concrete example**.
- After reading this illustrative example (and perhaps the **Introduction**), the readers should have a good understanding of the approach.
- *The goal is to have the readers (or reviewers) make up their mind whether they like or do not like the proposed work after reading this part.*
- A good place to add some **overview figure** highlighting the architecture and main components of the technique or tool.
- Can also summarize background or important concepts in this section.
- Typically contains:
 - An architecture or overview figure of the approach (e.g., Fig 1 in the [Dynaplex paper](#))
 - A motivation or illustration example
 - Sec 2.1: Show how [Dynaplex](#) works at high level on an input program, what results does it produce.
 - Main definitions and background (optional)
 - Sec 2.2: General background on dynamic inference and specific features/properties of [Dynaplex](#)

• **Experimental** section (Section 4).

1. Some implementation details. E.g., how the technique is implemented, what tools or libraries are used, etc.
2. Research Questions (RQs). E.g., RQ1: How accurate is the technique? RQ2: How efficient is the technique? RQ3: How does it compare to existing approaches?
3. Benchmark programs: describe benchmarks used to answer the RQs, and why they are good benchmarks for this purpose.
4. Experiment Setup
 - Machine specs
 - How experiments were run (multiple times and report the mean, etc.)
5. Results and Analysis
 - For each RQ, present the results and analyze them. E.g., for RQ1, show the accuracy results and analyze them (e.g., why is it accurate or not accurate? What are the factors that affect accuracy? etc.)
 - Don't just focus on positive results. If there are negative results, report and analyze them as well. E.g., for benchmark X, the technique does not perform well because of reasons Y and Z.

• **Algorithm/Techniques** (Section 3)

• **Related Work** (Section 5)

- Break into multiple subsections if necessary
- After each subsection that talks about related work, have a new paragraph and talk about why my work is different/related to other work.

• **Conclusion** (Section 6).

- This should be the *conclusion* not the *summary*.
- Future work can also go here

- **Introduction** (Section 1).

- Write this last, because by the time I write the Intro, I would already have all needed info (e.g., so I can summarize the diff with state of the art, technical contributions, and results).
- Typically contains several main things:
 - What is the problem? Why is it important (i.e., if we can solve it, then what do we accomplish)?
 - Existing work: basically answering why it is challenging (i.e., why existing work is not sufficient)?
 - The proposed approach of this paper: how does it address the limitations of existing work listed above? What other benefits does it have?
 - Results: briefly summarize the results.

2.3. Rebuttal

1. **Thank you note:** Start with a thank you note to the reviewers for their time and feedback. Reviewing is a thankless job and reviewers often put in lots of time and effort to provide feedback. Even if the reviews are harsh and you hate them, **do it anyway**.

2.4. Presentation

1. **Title slide** with paper title, authors (highlight or underline the presenter's name), affiliations, conference or event name, date.
2. **Number your slides.** This is useful when people want to ask questions (or give feedback) on specific slides (e.g., "*on slide 10, you said ...*").
3. **Every slide takes approximately 1 minute.** Using this rule of thumb, you can estimate how many slides you need for your presentation.
4. **Avoid Technical Stuff.**
 - Avoid slides that are too technical (e.g., formulae, code snippets, algorithms, etc) unless absolutely necessary.
 - *Focus on motivation and high-level ideas.* The goal of the presentation is to get people interested into your work and talk to you and read your paper, which has more technical details.
5. [Optional] Use **LaTeX/Beamer** for slides. It's easier to update and maintain consistency with the paper.

3. Lab Resources

3.1. Web Links

- **Website:** roars.dev. This is the main place for lab news, announcements, etc.
- **Zoom** for meetings meet.roars.dev: general lab meetings.

3.2. Discord Server

1. **#stats**: our most active channel, where we discuss research ideas, projects, papers, and everything else related to research and the lab. This is the main channel for research discussions and announcements.
2. **#servers**: for all questions regarding lab servers, e.g., how to get an account, how to use the servers, etc. This is the main channel for server-related discussions and announcements.
3. **#general**: where we say hello to new members

3.3. Computer Server Specs

We have *four* servers (see [pictures](#)):

1. **Boba** 🍷: ideal for AI-based projects, e.g., generative LLMs
 - [NVIDIA DGX Spark](#)
 - CPU: NVidia GB10 Chip, 20 core Arm (10 Cortex-X925 + 10 Cortex-A725 Arm)
 - Unified Memory/Graphic: 128GB RAM
 - OS: NVIDIA DGX OS (based on Ubuntu LTS)
2. **Pizza** 🍕: ideal for projects that can leverage many cores and specialized GPU
 - Custom built
 - CPU: 32 cores, 64 threads, 3.6GHz (AMD Ryzen Threadripper PRO 5975WX)
 - GPU: 2x Nvidia RTX 4090 24GB RAM
 - Memory: 128GB RAM
 - OS: Debian Stable
3. **Sushi** 🍣: high-performance workstation
 - Lenovo ThinkStation P620
 - CPU: 64 cores, 128 threads, 4.3GHz (AMD Ryzen Threadripper PRO 3995WX)
 - GPU: Nvidia RTX 5090 24GB RAM
 - Memory: 128GB RAM
 - OS: Debian Stable
4. **Taco** 🌮: ideal for experiments that need fast CPU (but not lots of cores) and fast GPU
 - Alienware Aurora
 - CPU: 16 cores, 24 threads, 5.20GHz (Intel 12th Gen Core i9-12900KF)
 - GPU: Nvidia RTX 3080 Ti 12GB RAM
 - Memory: 64GB RAM
 - OS: Debian Stable

Warning: For Debian machines, **DO NOT** install anything other than Debian stable packages to these computers (e.g., **do not mix unstable or new repositories** to `apt` sources list). You can download and install anything in your home (local) directory. If in doubt, contact us.

3.3.1. Getting an Account

To get an account on the server, send a request and your **SSH public key** (for *password-less* log in).

3.3.2. Miscellaneous

1. You can use an IDE such as `VSCode` to do remote development on these servers (just connect to the server via the builtin remote login method in VSCode). You can also use other editors that support remote development (e.g., `Emacs` + `TRAMP`).
2. Do take advantage of the multicores these machines have. Most programming languages support multiprocessing (e.g., `Python`) and can significantly reduce program runtime.
3. Periodically check `top` and `htop` to make sure **you** are not taking too much CPU or memory.
4. Periodically check `/var/tmp` or `/tmp/` to make sure **you** are not taking too much space there.

3.3.3. Pictures



Figure 1: 🐟 Sushi



Figure 2: 🐟 Sushi Cooling System



Figure 3: 🍕 Pizza



Figure 4: 🍷 Boba

3.4. External Resources

3.4.1. VPN for Lab Servers

You will need VPN and Computing access to log into lab servers remotely. Send an email to cechelp@gmu.edu and also CC the PI.

Subject: VPN and Computing Access (Mesa AD Groups)

Dear CECHelp,

I am writing to request that my NetID (**your_user_name**) be added to the Active Directory groups listed below.

Groups requested for addition:

- CN=VS-GRA,OU=HelpDeskManagement,OU=VolgenauSchool...
- CN=VS-SR-VPN,OU=HelpDeskManagement,OU=VolgenauSchool...

I have copied my advisor/supervisor, Professor Thanhvu Nguyen, on this email should you require confirmation that this access is necessary.

Thank you for your time and assistance with this.

3.4.2. Connecting to Wifi eduroam

[ITS Eduroam Documentation](#)

Linux Setup:

1. Select **eduroam** from the list of available wireless networks
2. Configure the following settings:
 - Wi-Fi security: **WPA & WPA2 Enterprise**
 - Authentication: **Protected EAP (PEAP)**
 - Anonymous identity: leave blank
 - Domain: **gmu.edu**
 - Check **No CA certificate is required**
 - PEAP version: **Automatic**
 - Inner Authentication: **MSCHAPv2**
 - Username: Your full George Mason email (e.g., **netid@gmu.edu**)
 - Password: Your Patriot Password
3. Click **Connect** to establish connection

4. Conference Registration and Travel

4.1. Reimbursement Process

4.1.1. General Process

- **PI Pay:** I can pay for your **conference registration** and **airline** in advance using the lab's **P-card**. Usually we do this together for efficiency.
- **You Pay (and get reimbursed):** Everything else (e.g., hotel, food, local transport) you pay yourself (e.g., using your own credit card) and then submit reimbursement requests to the university.
 - You need to save all receipts for reimbursement (except for meals, which is per diem see below [Section 4.1.2](#)).
 - You need to do this after you return from the conference and within 60 (?) days of the expense date.

4.1.2. Meals

1. **Per Diem:** We are paid per diem for meals when we travel for conferences. This means you don't need to save or submit receipts for meals. If you don't eat anything (do not do this)!, you will still get the per diem amount.
2. The per diem rates depend on the city we are visiting and are determined by the US General Services Administration (GSA) website: <https://www.gsa.gov/travel/plan-book/per-diem-rates>. When you submit reimbursement requests for meals, you will select the location and the system will automatically calculate the per diem amount for you.
3. The first day and the last day of travel, they will not pay the full per diem rate because you are not there for the full day.

5. Comprehensive Exam and Defense

5.1. Defense

Chair:

- In the beginning
 - Once all the committee members are present, chair will ask the student to step out of the room (virtually) at the beginning of the meeting.
 - Chair then give a brief summary of the student's progress in the PhD program, talk about their published and under-review papers, and answer any questions the committee may have.
 - Chair then call the student back and give the presentation.
- End of the meeting
 - the student briefly step out so that the committee can discuss the decision
 - Then the student can be invited and informed about the decision.

6. Contact information

1. Lab website: <https://roars.dev>
2. Email: info@roars.dev
3. Address (PI's office):

George Mason University
Nguyen Engineering Building #4430
4400 University Drive
Fairfax, VA 22030

4. Address (package delivery, e.g., from Amazon)

4511 Patriot CIR
Nguyen Engineering STE 4300
Fairfax, VA 22030-4469
United States

G LOR

G.1 Asking

- If you want a letter from me, you should **ask** at least 2 weeks in advance (ideally a month).
 - **DO NOT** put my name in the application system without explicitly asking me first.
- If I don't know you well enough, I will let you know and advise you to find someone else. But if you insist, I will still write you a letter.
- You need to **waive your right** to view the letter. It is a confidential evaluation, and I will not write one if you insist on seeing what I write.
- You should provide me your CV and, if you'd like, your SOP, so I can determine whether I could complement some of the things you've said.
 - Feel free to provide me with any other information that you think is relevant, e.g., your grades, projects, research experience, awards, etc. I might not use them if I don't think they are relevant, but they might help if I miss something.

G.2 Writing

- I **will not ask you to draft a letter** for me. My reputation is on the line, and I do not want a student (or AI) writing a letter on my behalf.
- I **do not write negative letters**, i.e., I don't say bad things about you (even if you're bad). However, if I don't know you well then a neutral, short, or generic letter will not help your case given the competitiveness.
- It takes me about **an hour** for a strong letter for someone I know well, and about 10 minutes for a letter for someone I don't know well.
- In all letters, at the end I will include a short paragraph about **an area you need to improve**, regardless of whether you're super strong or weak. But I will phrase it in such a way that the graduate study environment or a good advisor can help you overcome it. It makes the letter more complete and not just full of praise.
- I **do not customize** the letter for specific schools.
 - You customize your SOP to explain why you want to apply to school X and work with professor Y. I do not need to explain why you want to go to school X or why you would be a good fit for professor Y. Note that if I know you and Y very well, then I might send a letter directly to Y to mention you.
- My letter will have the university logo and my (digital) signature.
 - However, I should note that when I read a LoR, I do not pay attention to whether it has a logo or signature.

G.3 Sending

- After you put my email into the application system, it will send me a request email with a unique URL to go to. It will also give me some deadline that is *likely* not the same as yours—usually later.
 - In some cases the systems don't even give me a deadline (various LoR request examples are shown in my *Demystify* book, e.g., Section 3.2.3: <https://github.com/dynaroars/phd-cs-us/>).
- I **do not mind if the student is anxious and sends me multiple reminders**. I will likely not respond to them. But I am not bothered by reminders. Over-communication is better than under.
- Most systems simply ask me to upload the letter—though a few have short questions like comparing the applicant to undergrads or grad students. As an adcom reviewer, I don't really pay attention to these comparisons—only the letter matters.

- I usually send my letter (in PDF) in **batch mode**, e.g., I just sent out 10+ letters all at once and cleared my inbox of these requests.
 - For each request, it took me less than 30 seconds to upload the PDF and hit `submit`. So do not worry that it is time-consuming or a burden; it's not.
- After I send my batch, I'll let you know I just sent everything and ask if I missed any.
 - I would appreciate if you share an online spreadsheet showing what schools you have asked me to write letters for so I know what to expect (and if I miss something).

G.4 Updating

- Let me know your progress and especially your outcomes, e.g., interviews, offers, and where you eventually go to. I'd love to hear these updates from you.
- This is a common courtesy. Though I probably won't remember or expect you to do this. However, if you do this and then years later reach out for another letter, I might not be able to help you.